

1 Connect to the TurtleBot3

On your laptop (BlueDevel, RedDevel, etc.)

```
ping bluebot
```

If you get replies → OK.

Stop with **CTRL+C**.

SSH into the robot

```
ssh -l obv bluebot
```

Password:

```
turtle01
```

You are now inside the robot's Raspberry Pi (SBC).

2 Start the TurtleBot3 bringup

Inside the robot (SSH shell):

```
ros2 launch turtlebot3_bringup robot.launch.py
```

Leave this running.

Open a **second terminal** on your laptop.

3 Create your ROS2 workspace (on your laptop)

If you don't have a workspace yet:

```
mkdir -p ~/ros2_ws/src
cd ~/ros2_ws
colcon build
```

Source it:

```
source install/setup.bash
```

4 Create Lab1 package

Inside your workspace:

```
cd ~/ros2_ws/src
ros2 pkg create lab1pkg --build-type ament_python --dependencies rclpy geometry_msgs sensor_msgs
```

This creates:

```
lab1pkg/
  setup.py
  package.xml
  lab1pkg/
    __init__.py
```

5 Create your Python scripts

Go into the package:

```
cd ~/ros2_ws/src/lab1pkg/lab1pkg
```

Create **stop_turtle.py**:

```
nano stop_turtle.py
```

Paste this minimal working template:

```
import rclpy
from rclpy.node import Node
from geometry_msgs.msg import Twist
import time

class StopTurtle(Node):
    def __init__(self):
        super().__init__('stop_turtle_node')
        self.pub = self.create_publisher(Twist, '/cmd_vel', 10)
        self.get_logger().info("StopTurtle started")

        cmd = Twist()
        cmd.linear.x = 0.0
        cmd.angular.z = 0.0

        for _ in range(10):
            self.pub.publish(cmd)
            time.sleep(0.2)

def main(args=None):
    rclpy.init(args=args)
    node = StopTurtle()
    rclpy.shutdown()

if __name__ == '__main__':
    main()
```

Save: **CTRL+O**, ENTER

Exit: **CTRL+X**

Add executable to **setup.py**

Open:

```
nano ~/ros2_ws/src/lab1pkg/setup.py
```

Add inside `entry_points` :

```
'console_scripts': [  
    'stop_turtlebot3 = lab1pkg.stop_turtle:main',  
],
```

Save & exit.

Build the workspace

```
cd ~/ros2_ws  
colcon build  
source install/setup.bash
```

Run your node on the robot

On your laptop (NOT inside SSH):

```
ros2 run lab1pkg stop_turtlebot3
```

This publishes zero velocity to `/cmd_vel` and stops the robot.